

2

Linear Algebra

Some basic linear algebra facts will be reviewed in this chapter. The detailed treatment of this topic can be found in the references listed at the end of the chapter. Hence we shall omit most proofs and provide proofs only for those results that either cannot be easily found in the standard linear algebra textbooks or are insightful to the understanding of some related problems. We then treat a special class of matrix dilation problems which will be used in Chapters 8 and 17; however, most of the results presented in this book can be understood without the knowledge of the matrix dilation theory.

2.1 Linear Subspaces

Let $\mathbb{R}$ denote the real scalar field and $\mathbb{C}$ the complex scalar field. For the interest of this chapter, let $\mathbb{F}$ be either $\mathbb{R}$ or $\mathbb{C}$ and let $\mathbb{F}^n$ be the vector space over $\mathbb{F}$, i.e., $\mathbb{F}^n$ is either $\mathbb{R}^n$ or $\mathbb{C}^n$. Now let $x_1, x_2, \dots, x_k \in \mathbb{F}^n$. Then an element of the form $\alpha_1 x_1 + \dots + \alpha_k x_k$ with $\alpha_i \in \mathbb{F}$ is a *linear combination* over $\mathbb{F}$ of $x_1, \dots, x_k$. The set of all linear combinations of $x_1, x_2, \dots, x_k \in \mathbb{F}^n$ is a subspace called the *span* of $x_1, x_2, \dots, x_k$, denoted by

$$\text{span}\{x_1, x_2, \dots, x_k\} := \{x = \alpha_1 x_1 + \dots + \alpha_k x_k : \alpha_i \in \mathbb{F}\}.$$

A set of vectors $x_1, x_2, \dots, x_k \in \mathbb{F}^n$ are said to be *linearly dependent* over $\mathbb{F}$ if there exists $\alpha_1, \dots, \alpha_k \in \mathbb{F}$ not all zero such that $\alpha_1 x_1 + \dots + \alpha_k x_k = 0$; otherwise they are said to be *linearly independent*.

Let S be a subspace of $\mathbb{F}^n$, then a set of vectors $\{x_1, x_2, \dots, x_k\} \in S$ is called a *basis* for S if $x_1, x_2, \dots, x_k$ are linearly independent and $S = \text{span}\{x_1, x_2, \dots, x_k\}$. However,

such a basis for a subspace S is not unique but all bases for S have the same number of elements. This number is called the *dimension* of S , denoted by $\dim(S)$.

A set of vectors $\{x_1, x_2, \dots, x_k\}$ in $\mathbb{F}^n$ are mutually *orthogonal* if $x_i^* x_j = 0$ for all $i \neq j$ and *orthonormal* if $x_i^* x_j = \delta_{ij}$, where the superscript $*$ denotes complex conjugate transpose and δ_{ij} is the Kronecker delta function with $\delta_{ij} = 1$ for $i = j$ and $\delta_{ij} = 0$ for $i \neq j$. More generally, a collection of subspaces $S_1, S_2, \dots, S_k$ of $\mathbb{F}^n$ are mutually orthogonal if $x^* y = 0$ whenever $x \in S_i$ and $y \in S_j$ for $i \neq j$.

The *orthogonal complement* of a subspace $S \subset \mathbb{F}^n$ is defined by

$$S^\perp := \{y \in \mathbb{F}^n : y^* x = 0 \text{ for all } x \in S\}.$$

We call a set of vectors $\{u_1, u_2, \dots, u_k\}$ an *orthonormal basis* for a subspace $S \in \mathbb{F}^n$ if they form a basis of S and are orthonormal. It is always possible to extend such a basis to a full orthonormal basis $\{u_1, u_2, \dots, u_n\}$ for $\mathbb{F}^n$. Note that in this case

$$S^\perp = \text{span}\{u_{k+1}, \dots, u_n\},$$

and $\{u_{k+1}, \dots, u_n\}$ is called an *orthonormal completion* of $\{u_1, u_2, \dots, u_k\}$.

Let $A \in \mathbb{F}^{m \times n}$ be a linear transformation from $\mathbb{F}^n$ to $\mathbb{F}^m$, i.e.,

$$A : \mathbb{F}^n \mapsto \mathbb{F}^m.$$

(Note that a vector $x \in \mathbb{F}^m$ can also be viewed as a linear transformation from $\mathbb{F}$ to $\mathbb{F}^m$, hence anything said for the general matrix case is also true for the vector case.) Then the *kernel* or *null space* of the linear transformation A is defined by

$$\text{Ker} A = N(A) := \{x \in \mathbb{F}^n : Ax = 0\},$$

and the *image* or *range* of A is

$$\text{Im} A = R(A) := \{y \in \mathbb{F}^m : y = Ax, x \in \mathbb{F}^n\}.$$

It is clear that $\text{Ker} A$ is a subspace of $\mathbb{F}^n$ and $\text{Im} A$ is a subspace of $\mathbb{F}^m$. Moreover, it can be easily seen that $\dim(\text{Ker} A) + \dim(\text{Im} A) = n$ and $\dim(\text{Im} A) = \dim(\text{Ker} A)^\perp$. Note that $(\text{Ker} A)^\perp$ is a subspace of $\mathbb{F}^n$.

Let $a_i, i = 1, 2, \dots, n$ denote the columns of a matrix $A \in \mathbb{F}^{m \times n}$, then

$$\text{Im} A = \text{span}\{a_1, a_2, \dots, a_n\}.$$

The rank of a matrix A is defined by

$$\text{rank}(A) = \dim(\text{Im} A).$$

It is a fact that $\text{rank}(A) = \text{rank}(A^*)$, and thus the rank of a matrix equals the maximal number of independent rows or columns. A matrix $A \in \mathbb{F}^{m \times n}$ is said to have *full row rank* if $m \leq n$ and $\text{rank}(A) = m$. Dually, it is said to have *full column rank* if $n \leq m$ and $\text{rank}(A) = n$. A full rank square matrix is called a *nonsingular matrix*. It is easy

to see that $\text{rank}(A) = \text{rank}(AT) = \text{rank}(PA)$ if T and P are nonsingular matrices with appropriate dimensions.

A square matrix $U \in \mathbb{F}^{n \times n}$ whose columns form an orthonormal basis for $\mathbb{F}^n$ is called an *unitary matrix* (or *orthogonal matrix* if $\mathbb{F} = \mathbb{R}$), and it satisfies $U^*U = I = UU^*$. The following lemma is useful.

Lemma 2.1 *Let $D = [d_1 \ \dots \ d_k] \in \mathbb{F}^{n \times k}$ ($n > k$) be such that $D^*D = I$, so $d_i, i = 1, 2, \dots, k$ are orthonormal. Then there exists a matrix $D_\perp \in \mathbb{F}^{n \times (n-k)}$ such that $[D \ D_\perp]$ is a unitary matrix. Furthermore, the columns of $D_\perp$, $d_i, i = k+1, \dots, n$, form an orthonormal completion of $\{d_1, d_2, \dots, d_k\}$.*

The following results are standard:

Lemma 2.2 *Consider the linear equation*

$$AX = B$$

where $A \in \mathbb{F}^{n \times l}$ and $B \in \mathbb{F}^{n \times m}$ are given matrices. Then the following statements are equivalent:

- (i) *there exists a solution $X \in \mathbb{F}^{l \times m}$.*
- (ii) *the columns of $B \in \text{Im}A$.*
- (iii) *$\text{rank}[A \ B] = \text{rank}(A)$.*
- (iv) *$\text{Ker}(A^*) \subset \text{Ker}(B^*)$.*

Furthermore, the solution, if it exists, is unique if and only if A has full column rank.

The following lemma concerns the rank of the product of two matrices.

Lemma 2.3 (Sylvester's inequality) *Let $A \in \mathbb{F}^{m \times n}$ and $B \in \mathbb{F}^{n \times k}$. Then*

$$\text{rank}(A) + \text{rank}(B) - n \leq \text{rank}(AB) \leq \min\{\text{rank}(A), \text{rank}(B)\}.$$

For simplicity, a matrix M with m_{ij} as its i -th row and j -th column's element will sometimes be denoted as $M = [m_{ij}]$ in this book. We will mostly use I as above to denote an identity matrix with compatible dimensions, but from time to time, we will use I_n to emphasize that it is an $n \times n$ identity matrix.

Now let $A = [a_{ij}] \in \mathbb{C}^{n \times n}$, then the *trace* of A is defined as

$$\text{Trace}(A) := \sum_{i=1}^n a_{ii}.$$

Trace has the following properties:

$$\text{Trace}(\alpha A) = \alpha \text{Trace}(A), \quad \forall \alpha \in \mathbb{C}, \ A \in \mathbb{C}^{n \times n}$$

$$\text{Trace}(A + B) = \text{Trace}(A) + \text{Trace}(B), \quad \forall A, B \in \mathbb{C}^{n \times n}$$

$$\text{Trace}(AB) = \text{Trace}(BA), \quad \forall A \in \mathbb{C}^{n \times m}, \ B \in \mathbb{C}^{m \times n}.$$

2.2 Eigenvalues and Eigenvectors

Let $A \in \mathbb{C}^{n \times n}$, then the *eigenvalues* of A are the n roots of its characteristic polynomial $p(\lambda) = \det(\lambda I - A)$. This set of roots is called the *spectrum* of A and is denoted by $\sigma(A)$ (not to be confused with singular values defined later). That is, $\sigma(A) := \{\lambda_1, \lambda_2, \dots, \lambda_n\}$ if λ_i is a root of $p(\lambda)$. The maximal modulus of the eigenvalues is called the *spectral radius*, denoted by

$$\rho(A) := \max_{1 \leq i \leq n} |\lambda_i|$$

where, as usual, $|\cdot|$ denotes the magnitude.

If $\lambda \in \sigma(A)$ then any nonzero vector $x \in \mathbb{C}^n$ that satisfies

$$Ax = \lambda x$$

is referred to as a *right eigenvector* of A . Dually, a nonzero vector y is called a *left eigenvector* of A if

$$y^* A = \lambda y^*.$$

It is a well known (but nontrivial) fact in linear algebra that any complex matrix admits a *Jordan Canonical* representation:

Theorem 2.4 *For any square complex matrix $A \in \mathbb{C}^{n \times n}$, there exists a nonsingular matrix T such that*

$$A = T J T^{-1}$$

where

$$\begin{aligned} J &= \text{diag}\{J_1, J_2, \dots, J_l\} \\ J_i &= \text{diag}\{J_{i1}, J_{i2}, \dots, J_{im_i}\} \\ J_{ij} &= \begin{bmatrix} \lambda_i & 1 & & & \\ & \lambda_i & 1 & & \\ & & \ddots & \ddots & \\ & & & \lambda_i & 1 \\ & & & & \lambda_i \end{bmatrix} \in \mathbb{C}^{n_{ij} \times n_{ij}} \end{aligned}$$

with $\sum_{i=1}^l \sum_{j=1}^{m_i} n_{ij} = n$, and with $\{\lambda_i : i = 1, \dots, l\}$ as the distinct eigenvalues of A .

The transformation T has the following form:

$$\begin{aligned} T &= [T_1 \quad T_2 \quad \dots \quad T_l] \\ T_i &= [T_{i1} \quad T_{i2} \quad \dots \quad T_{im_i}] \\ T_{ij} &= [t_{ij1} \quad t_{ij2} \quad \dots \quad t_{ijn_{ij}}] \end{aligned}$$

where t_{ij1} are the eigenvectors of A ,

$$A t_{ij1} = \lambda_i t_{ij1},$$

and $t_{ijk} \neq 0$ defined by the following linear equations for $k \geq 2$

$$(A - \lambda_i I)t_{ijk} = t_{ij(k-1)}$$

are called the *generalized eigenvectors* of A . For a given integer $q \leq n_{ij}$, the generalized eigenvectors t_{ijl} , $\forall l < q$, are called the *lower rank generalized eigenvectors* of t_{ijq} .

Definition 2.1 A square matrix $A \in \mathbb{R}^{n \times n}$ is called *cyclic* if the Jordan canonical form of A has one and only one Jordan block associated with each distinct eigenvalue.

More specifically, a matrix A is cyclic if its Jordan form has $m_i = 1, i = 1, \dots, l$. Clearly, a square matrix A with all distinct eigenvalues is cyclic and can be diagonalized:

$$A \begin{bmatrix} x_1 & x_2 & \cdots & x_n \end{bmatrix} = \begin{bmatrix} x_1 & x_2 & \cdots & x_n \end{bmatrix} \begin{bmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_n \end{bmatrix}.$$

In this case, A has the following *spectral decomposition*:

$$A = \sum_{i=1}^n \lambda_i x_i y_i^*$$

where $y_i \in \mathbb{C}^n$ is given by

$$\begin{bmatrix} y_1^* \\ y_2^* \\ \vdots \\ y_n^* \end{bmatrix} = \begin{bmatrix} x_1 & x_2 & \cdots & x_n \end{bmatrix}^{-1}.$$

In general, eigenvalues need not be real, and neither do their corresponding eigenvectors. However, if A is real and λ is a real eigenvalue of A , then there is a real eigenvector corresponding to λ . In the case that all eigenvalues of a matrix A are real¹, we will denote $\lambda_{max}(A)$ for the largest eigenvalue of A and $\lambda_{min}(A)$ for the smallest eigenvalue. In particular, if A is a Hermitian matrix, then there exist a unitary matrix U and a real diagonal matrix Λ such that $A = U\Lambda U^*$, where the diagonal elements of Λ are the eigenvalues of A and the columns of U are the eigenvectors of A .

The following theorem is useful in linear system theory.

Theorem 2.5 (Cayley-Hamilton) Let $A \in \mathbb{C}^{n \times n}$ and denote

$$\det(\lambda I - A) = \lambda^n + a_1 \lambda^{n-1} + \cdots + a_n.$$

Then

$$A^n + a_1 A^{n-1} + \cdots + a_n I = 0.$$

¹For example, this is the case if A is *Hermitian*, i.e., $A = A^*$.

This is obvious if A has distinct eigenvalues. Since

$$A^n + a_1 A^{n-1} + \cdots + a_n I = T^{-1} \text{diag} \{ \dots, \lambda_i^n + a_1 \lambda_i^{n-1} + \cdots + a_n, \dots \} T = 0,$$

and λ_i is an eigenvalue of A . The proof for the general case follows from the following lemma.

Lemma 2.6 *Let $A \in \mathbb{C}^{n \times n}$. Then*

$$(\lambda I - A)^{-1} = \frac{1}{\det(\lambda I - A)} (R_1 \lambda^{n-1} + R_2 \lambda^{n-2} + \cdots + R_n)$$

and

$$\det(\lambda I - A) = \lambda^n + a_1 \lambda^{n-1} + \cdots + a_n$$

where a_i and R_i can be computed from the following recursive formulas:

$$\begin{array}{ll} a_1 &= -\text{Trace } A & R_1 &= I \\ a_2 &= -\frac{1}{2} \text{Trace}(R_2 A) & R_2 &= R_1 A + a_1 I \\ &\vdots & &\vdots \\ a_{n-1} &= -\frac{1}{n-1} \text{Trace}(R_{n-1} A) & R_n &= R_{n-1} A + a_{n-1} I \\ a_n &= -\frac{1}{n} \text{Trace}(R_n A) & 0 &= R_n A + a_n I. \end{array}$$

The proof is left to the reader as an exercise. Note that the Cayley-Hamilton Theorem follows from the fact that

$$0 = R_n A + a_n I = A^n + a_1 A^{n-1} + \cdots + a_n I.$$

2.3 Matrix Inversion Formulas

Let A be a square matrix partitioned as follows

$$A := \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}$$

where A_{11} and A_{22} are also square matrices. Now suppose A_{11} is nonsingular, then A has the following decomposition:

$$\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} = \begin{bmatrix} I & 0 \\ A_{21} A_{11}^{-1} & I \end{bmatrix} \begin{bmatrix} A_{11} & 0 \\ 0 & \Delta \end{bmatrix} \begin{bmatrix} I & A_{11}^{-1} A_{12} \\ 0 & I \end{bmatrix}$$

with $\Delta := A_{22} - A_{21} A_{11}^{-1} A_{12}$, and A is nonsingular iff Δ is nonsingular.

Dually, if A_{22} is nonsingular, then

$$\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} = \begin{bmatrix} I & A_{12} A_{22}^{-1} \\ 0 & I \end{bmatrix} \begin{bmatrix} \overline{\Delta} & 0 \\ 0 & A_{22} \end{bmatrix} \begin{bmatrix} I & 0 \\ A_{22}^{-1} A_{21} & I \end{bmatrix}$$

with $\overline{\Delta} := A_{11} - A_{12}A_{22}^{-1}A_{21}$, and A is nonsingular iff $\overline{\Delta}$ is nonsingular. The matrix Δ ($\overline{\Delta}$) is called the *Schur complement* of A_{11} (A_{22}) in A .

Moreover, if A is nonsingular, then

$$\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}^{-1} = \begin{bmatrix} A_{11}^{-1} + A_{11}^{-1}A_{12}\Delta^{-1}A_{21}A_{11}^{-1} & -A_{11}^{-1}A_{12}\Delta^{-1} \\ -\Delta^{-1}A_{21}A_{11}^{-1} & \Delta^{-1} \end{bmatrix}$$

and

$$\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}^{-1} = \begin{bmatrix} \overline{\Delta}^{-1} & -\overline{\Delta}^{-1}A_{12}A_{22}^{-1} \\ -A_{22}^{-1}A_{21}\overline{\Delta}^{-1} & A_{22}^{-1} + A_{22}^{-1}A_{21}\overline{\Delta}^{-1}A_{12}A_{22}^{-1} \end{bmatrix}.$$

The above matrix inversion formulas are particularly simple if A is block triangular:

$$\begin{aligned} \begin{bmatrix} A_{11} & 0 \\ A_{21} & A_{22} \end{bmatrix}^{-1} &= \begin{bmatrix} A_{11}^{-1} & 0 \\ -A_{22}^{-1}A_{21}A_{11}^{-1} & A_{22}^{-1} \end{bmatrix} \\ \begin{bmatrix} A_{11} & A_{12} \\ 0 & A_{22} \end{bmatrix}^{-1} &= \begin{bmatrix} A_{11}^{-1} & -A_{11}^{-1}A_{12}A_{22}^{-1} \\ 0 & A_{22}^{-1} \end{bmatrix}. \end{aligned}$$

The following identity is also very useful. Suppose A_{11} and A_{22} are both nonsingular matrices, then

$$(A_{11} - A_{12}A_{22}^{-1}A_{21})^{-1} = A_{11}^{-1} + A_{11}^{-1}A_{12}(A_{22} - A_{21}A_{11}^{-1}A_{12})^{-1}A_{21}A_{11}^{-1}.$$

As a consequence of the matrix decomposition formulas mentioned above, we can calculate the determinant of a matrix by using its sub-matrices. Suppose A_{11} is nonsingular, then

$$\det A = \det A_{11} \det(A_{22} - A_{21}A_{11}^{-1}A_{12}).$$

On the other hand, if A_{22} is nonsingular, then

$$\det A = \det A_{22} \det(A_{11} - A_{12}A_{22}^{-1}A_{21}).$$

In particular, for any $B \in \mathbb{C}^{m \times n}$ and $C \in \mathbb{C}^{n \times m}$, we have

$$\det \begin{bmatrix} I_m & B \\ -C & I_n \end{bmatrix} = \det(I_n + CB) = \det(I_m + BC)$$

and for $x, y \in \mathbb{C}^n$

$$\det(I_n + xy^*) = 1 + y^*x.$$

2.4 Matrix Calculus

Let $X = [x_{ij}] \in \mathbb{C}^{m \times n}$ be a real or complex matrix and $F(X) \in \mathbb{C}$ be a scalar real or complex function of X ; then the derivative of $F(X)$ with respect to X is defined as

$$\frac{\partial}{\partial X} F(X) := \left[\frac{\partial}{\partial x_{ij}} F(X) \right].$$

Let A and B be constant complex matrices with compatible dimensions. Then the following is a list of formulas for the derivatives²:

$$\begin{aligned} \frac{\partial}{\partial X} \text{Trace}\{AXB\} &= A^T B^T \\ \frac{\partial}{\partial X} \text{Trace}\{AX^T B\} &= BA \\ \frac{\partial}{\partial X} \text{Trace}\{AXBX\} &= A^T X^T B^T + B^T X^T A^T \\ \frac{\partial}{\partial X} \text{Trace}\{AXBX^T\} &= A^T X B^T + AXB \\ \frac{\partial}{\partial X} \text{Trace}\{X^k\} &= k(X^{k-1})^T \\ \frac{\partial}{\partial X} \text{Trace}\{AX^k\} &= \left(\sum_{i=0}^{k-1} X^i A X^{k-i-1} \right)^T \\ \frac{\partial}{\partial X} \text{Trace}\{AX^{-1}B\} &= -(X^{-1}BA X^{-1})^T \\ \frac{\partial}{\partial X} \log \det X &= (X^T)^{-1} \\ \frac{\partial}{\partial X} \det X^T &= \frac{\partial}{\partial X} \det X = (\det X)(X^T)^{-1} \\ \frac{\partial}{\partial X} \det\{X^k\} &= k(\det X^k)(X^T)^{-1}. \end{aligned}$$

And finally, the derivative of a matrix $A(\alpha) \in \mathbb{C}^{m \times n}$ with respect to a scalar $\alpha \in \mathbb{C}$ is defined as

$$\frac{dA}{d\alpha} := \left[\frac{da_{ij}}{d\alpha} \right]$$

so that all the rules applicable to a scalar function also apply here. In particular, we have

$$\begin{aligned} \frac{d(AB)}{d\alpha} &= \frac{dA}{d\alpha} B + A \frac{dB}{d\alpha} \\ \frac{dA^{-1}}{d\alpha} &= -A^{-1} \frac{dA}{d\alpha} A^{-1}. \end{aligned}$$

²Note that transpose rather than complex conjugate transpose should be used in the list even if the involved matrices are complex matrices.

2.5 Kronecker Product and Kronecker Sum

Let $A \in \mathbb{C}^{m \times n}$ and $B \in \mathbb{C}^{p \times q}$, then the *Kronecker product* of A and B is defined as

$$A \otimes B := \begin{bmatrix} a_{11}B & a_{12}B & \cdots & a_{1n}B \\ a_{21}B & a_{22}B & \cdots & a_{2n}B \\ \vdots & \vdots & & \vdots \\ a_{m1}B & a_{m2}B & \cdots & a_{mn}B \end{bmatrix} \in \mathbb{C}^{mp \times nq}.$$

Furthermore, if the matrices A and B are square and $A \in \mathbb{C}^{n \times n}$ and $B \in \mathbb{C}^{m \times m}$ then the *Kronecker sum* of A and B is defined as

$$A \oplus B := (A \otimes I_m) + (I_n \otimes B) \in \mathbb{C}^{nm \times nm}.$$

Let $X \in \mathbb{C}^{m \times n}$ and let $\text{vec}(X)$ denote the vector formed by stacking the columns of X into one long vector:

$$\text{vec}(X) := \begin{bmatrix} x_{11} \\ x_{21} \\ \vdots \\ x_{m1} \\ x_{12} \\ x_{22} \\ \vdots \\ x_{1n} \\ x_{2n} \\ \vdots \\ x_{mn} \end{bmatrix}.$$

Then for any matrices $A \in \mathbb{C}^{k \times m}$, $B \in \mathbb{C}^{n \times l}$, and $X \in \mathbb{C}^{m \times n}$, we have

$$\text{vec}(AXB) = (B^T \otimes A)\text{vec}(X).$$

Consequently, if $k = m$ and $l = n$, then

$$\text{vec}(AX + XB) = (B^T \oplus A)\text{vec}(X).$$

Let $A \in \mathbb{C}^{n \times n}$ and $B \in \mathbb{C}^{m \times m}$, and let $\{\lambda_i, i = 1, \dots, n\}$ be the eigenvalues of A and $\{\mu_j, j = 1, \dots, m\}$ be the eigenvalues of B . Then we have the following properties:

- The eigenvalues of $A \otimes B$ are the mn numbers $\lambda_i \mu_j, i = 1, 2, \dots, n, j = 1, 2, \dots, m$.
- The eigenvalues of $A \oplus B = (A \otimes I_m) + (I_n \otimes B)$ are the mn numbers $\lambda_i + \mu_j, i = 1, 2, \dots, n, j = 1, 2, \dots, m$.

- Let $\{x_i, i = 1, \dots, n\}$ be the eigenvectors of A and let $\{y_j, j = 1, \dots, m\}$ be the eigenvectors of B . Then the eigenvectors of $A \otimes B$ and $A \oplus B$ correspond to the eigenvalues $\lambda_i \mu_j$ and $\lambda_i + \mu_j$ are $x_i \otimes y_j$.

Using these properties, we can show the following Lemma.

Lemma 2.7 *Consider the Sylvester equation*

$$AX + XB = C \quad (2.1)$$

where $A \in \mathbb{F}^{n \times n}$, $B \in \mathbb{F}^{m \times m}$, and $C \in \mathbb{F}^{n \times m}$ are given matrices. There exists a unique solution $X \in \mathbb{F}^{n \times m}$ if and only if $\lambda_i(A) + \lambda_j(B) \neq 0$, $\forall i = 1, 2, \dots, n$ and $j = 1, 2, \dots, m$.

In particular, if $B = A^*$, (2.1) is called the “Lyapunov Equation”; and the necessary and sufficient condition for the existence of a unique solution is that $\lambda_i(A) + \bar{\lambda}_j(A) \neq 0$, $\forall i, j = 1, 2, \dots, n$

Proof. Equation (2.1) can be written as a linear matrix equation by using the Kronecker product:

$$(B^T \oplus A)\text{vec}(X) = \text{vec}(C).$$

Now this equation has a unique solution iff $B^T \oplus A$ is nonsingular. Since the eigenvalues of $B^T \oplus A$ have the form of $\lambda_i(A) + \lambda_j(B^T) = \lambda_i(A) + \lambda_j(B)$, the conclusion follows. $\square$

The properties of the Lyapunov equations will be studied in more detail in the next chapter.

2.6 Invariant Subspaces

Let $A : \mathbb{C}^n \mapsto \mathbb{C}^n$ be a linear transformation, λ be an eigenvalue of A , and x be a corresponding eigenvector, respectively. Then $Ax = \lambda x$ and $A(\alpha x) = \lambda(\alpha x)$ for any $\alpha \in \mathbb{C}$. Clearly, the eigenvector x defines a one-dimensional subspace that is invariant with respect to pre-multiplication by A since $A^k x = \lambda^k x, \forall k$. In general, a subspace $S \subset \mathbb{C}^n$ is called *invariant* for the transformation A , or *A-invariant*, if $Ax \in S$ for every $x \in S$. In other words, that S is invariant for A means that the image of S under A is contained in S : $AS \subset S$. For example, $\{0\}$, $\mathbb{C}^n$, $\text{Ker} A$, and $\text{Im} A$ are all A -invariant subspaces.

As a generalization of the one dimensional invariant subspace induced by an eigenvector, let $\lambda_1, \dots, \lambda_k$ be eigenvalues of A (not necessarily distinct), and let x_i be the corresponding eigenvectors and the generalized eigenvectors. Then $S = \text{span}\{x_1, \dots, x_k\}$ is an A -invariant subspace provided that all the lower rank generalized eigenvectors are included. More specifically, let $\lambda_1 = \lambda_2 = \dots = \lambda_l$ be eigenvalues of A , and

let $x_1, x_2, \dots, x_l$ be the corresponding eigenvector and the generalized eigenvectors obtained through the following equations:

$$\begin{aligned} (A - \lambda_1 I)x_1 &= 0 \\ (A - \lambda_1 I)x_2 &= x_1 \\ &\vdots \\ (A - \lambda_1 I)x_l &= x_{l-1}. \end{aligned}$$

Then a subspace S with $x_t \in S$ for some $t \leq l$ is an A -invariant subspace only if all lower rank eigenvectors and generalized eigenvectors of x_t are in S , i.e., $x_i \in S$, $\forall 1 \leq i \leq t$. This will be further illustrated in Example 2.1.

On the other hand, if S is a nontrivial subspace³ and is A -invariant, then there is $x \in S$ and λ such that $Ax = \lambda x$.

An A -invariant subspace $S \subset \mathbb{C}^n$ is called a *stable invariant subspace* if all the eigenvalues of A constrained to S have negative real parts. Stable invariant subspaces will play an important role in computing the stabilizing solutions to the algebraic Riccati equations in Chapter 13.

Example 2.1 Suppose a matrix A has the following Jordan canonical form

$$A \begin{bmatrix} x_1 & x_2 & x_3 & x_4 \end{bmatrix} = \begin{bmatrix} x_1 & x_2 & x_3 & x_4 \end{bmatrix} \begin{bmatrix} \lambda_1 & 1 & & \\ & \lambda_1 & & \\ & & \lambda_3 & \\ & & & \lambda_4 \end{bmatrix}$$

with $\text{Re}\lambda_1 < 0$, $\lambda_3 < 0$, and $\lambda_4 > 0$. Then it is easy to verify that

$$\begin{aligned} S_1 &= \text{span}\{x_1\} & S_{12} &= \text{span}\{x_1, x_2\} & S_{123} &= \text{span}\{x_1, x_2, x_3\} \\ S_3 &= \text{span}\{x_3\} & S_{13} &= \text{span}\{x_1, x_3\} & S_{124} &= \text{span}\{x_1, x_2, x_4\} \\ S_4 &= \text{span}\{x_4\} & S_{14} &= \text{span}\{x_1, x_4\} & S_{34} &= \text{span}\{x_3, x_4\} \end{aligned}$$

are all A -invariant subspaces. Moreover, S_1, S_3, S_{12}, S_{13} , and S_{123} are stable A -invariant subspaces. However, the subspaces $S_2 = \text{span}\{x_2\}$, $S_{23} = \text{span}\{x_2, x_3\}$, $S_{24} = \text{span}\{x_2, x_4\}$, and $S_{234} = \text{span}\{x_2, x_3, x_4\}$ are not A -invariant subspaces since the lower rank generalized eigenvector x_1 of x_2 is not in these subspaces. To illustrate, consider the subspace S_{23} . Then by definition, $Ax_2 \in S_{23}$ if it is an A -invariant subspace. Since

$$Ax_2 = \lambda x_2 + x_1,$$

$Ax_2 \in S_{23}$ would require that x_1 be a linear combination of x_2 and x_3 , but this is impossible since x_1 is independent of x_2 and x_3 . $\diamond$

³We will say subspace S is trivial if $S = \{0\}$.

2.7 Vector Norms and Matrix Norms

In this section, we will define vector and matrix norms. Let X be a vector space, a real-valued function $\|\cdot\|$ defined on X is said to be a *norm* on X if it satisfies the following properties:

- (i) $\|x\| \geq 0$ (positivity);
- (ii) $\|x\| = 0$ if and only if $x = 0$ (positive definiteness);
- (iii) $\|\alpha x\| = |\alpha| \|x\|$, for any scalar α (homogeneity);
- (iv) $\|x + y\| \leq \|x\| + \|y\|$ (triangle inequality)

for any $x \in X$ and $y \in X$. A function is said to be a *semi-norm* if it satisfies (i), (iii), and (iv) but not necessarily (ii).

Let $x \in \mathbb{C}^n$. Then we define the vector p -norm of x as

$$\|x\|_p := \left(\sum_{i=1}^n |x_i|^p \right)^{1/p}, \text{ for } 1 \leq p \leq \infty.$$

In particular, when $p = 1, 2, \infty$ we have

$$\|x\|_1 := \sum_{i=1}^n |x_i|;$$

$$\|x\|_2 := \sqrt{\sum_{i=1}^n |x_i|^2};$$

$$\|x\|_\infty := \max_{1 \leq i \leq n} |x_i|.$$

Clearly, norm is an abstraction and extension of our usual concept of length in 3-dimensional Euclidean space. So a norm of a vector is a measure of the vector “length”, for example $\|x\|_2$ is the Euclidean distance of the vector x from the origin. Similarly, we can introduce some kind of measure for a matrix.

Let $A = [a_{ij}] \in \mathbb{C}^{m \times n}$, then the matrix norm *induced* by a vector p -norm is defined as

$$\|A\|_p := \sup_{x \neq 0} \frac{\|Ax\|_p}{\|x\|_p}.$$

In particular, for $p = 1, 2, \infty$, the corresponding induced matrix norm can be computed as

$$\|A\|_1 = \max_{1 \leq j \leq n} \sum_{i=1}^m |a_{ij}| \quad (\text{column sum});$$

$$\|A\|_2 = \sqrt{\lambda_{\max}(A^*A)} ;$$

$$\|A\|_\infty = \max_{1 \leq i \leq m} \sum_{j=1}^n |a_{ij}| \quad (\text{row sum}) .$$

The matrix norms induced by vector p -norms are sometimes called *induced p -norms*. This is because $\|A\|_p$ is defined by or induced from a vector p -norm. In fact, A can be viewed as a mapping from a vector space $\mathbb{C}^n$ equipped with a vector norm $\|\cdot\|_p$ to another vector space $\mathbb{C}^m$ equipped with a vector norm $\|\cdot\|_p$. So from a system theoretical point of view, the induced norms have the interpretation of input/output amplification gains.

We shall adopt the following convention throughout the book for the vector and matrix norms unless specified otherwise: let $x \in \mathbb{C}^n$ and $A \in \mathbb{C}^{m \times n}$, then we shall denote the Euclidean 2-norm of x simply by

$$\|x\| := \|x\|_2$$

and the induced 2-norm of A by

$$\|A\| := \|A\|_2 .$$

The Euclidean 2-norm has some very nice properties:

Lemma 2.8 *Let $x \in \mathbb{F}^n$ and $y \in \mathbb{F}^m$.*

1. *Suppose $n \geq m$. Then $\|x\| = \|y\|$ iff there is a matrix $U \in \mathbb{F}^{n \times m}$ such that $x = Uy$ and $U^*U = I$.*
2. *Suppose $n = m$. Then $|x^*y| \leq \|x\| \|y\|$. Moreover, the equality holds iff $x = \alpha y$ for some $\alpha \in \mathbb{F}$ or $y = 0$.*
3. *$\|x\| \leq \|y\|$ iff there is a matrix $\Delta \in \mathbb{F}^{n \times m}$ with $\|\Delta\| \leq 1$ such that $x = \Delta y$. Furthermore, $\|x\| < \|y\|$ iff $\|\Delta\| < 1$.*
4. *$\|Ux\| = \|x\|$ for any appropriately dimensioned unitary matrices U .*

Another often used matrix norm is the so called *Frobenius norm*. It is defined as

$$\|A\|_F := \sqrt{\text{Trace}(A^*A)} = \sqrt{\sum_{i=1}^m \sum_{j=1}^n |a_{ij}|^2} .$$

However, the Frobenius norm is not an induced norm.

The following properties of matrix norms are easy to show:

Lemma 2.9 *Let A and B be any matrices with appropriate dimensions. Then*

1. $\rho(A) \leq \|A\|$ (This is also true for F norm and any induced matrix norm).
2. $\|AB\| \leq \|A\| \|B\|$. In particular, this gives $\|A^{-1}\| \geq \|A\|^{-1}$ if A is invertible. (This is also true for any induced matrix norm.)
3. $\|UAV\| = \|A\|$, and $\|UAV\|_F = \|A\|_F$, for any appropriately dimensioned unitary matrices U and V .
4. $\|AB\|_F \leq \|A\| \|B\|_F$ and $\|AB\|_F \leq \|B\| \|A\|_F$.

Note that although pre-multiplication or post-multiplication of a unitary matrix on a matrix does not change its induced 2-norm and F -norm, it does change its eigenvalues. For example, let

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}.$$

Then $\lambda_1(A) = 1, \lambda_2(A) = 0$. Now let

$$U = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix};$$

then U is a unitary matrix and

$$UA = \begin{bmatrix} \sqrt{2} & 0 \\ 0 & 0 \end{bmatrix}$$

with $\lambda_1(UA) = \sqrt{2}, \lambda_2(UA) = 0$. This property is useful in some matrix perturbation problems, particularly, in the computation of bounds for structured singular values which will be studied in Chapter 10.

Lemma 2.10 *Let A be a block partitioned matrix with*

$$A = \begin{bmatrix} A_{11} & A_{12} & \cdots & A_{1q} \\ A_{21} & A_{22} & \cdots & A_{2q} \\ \vdots & \vdots & & \vdots \\ A_{m1} & A_{m2} & \cdots & A_{mq} \end{bmatrix} =: [A_{ij}],$$

and let each A_{ij} be an appropriately dimensioned matrix. Then for any induced matrix p -norm

$$\|A\|_p \leq \left\| \begin{bmatrix} \|A_{11}\|_p & \|A_{12}\|_p & \cdots & \|A_{1q}\|_p \\ \|A_{21}\|_p & \|A_{22}\|_p & \cdots & \|A_{2q}\|_p \\ \vdots & \vdots & & \vdots \\ \|A_{m1}\|_p & \|A_{m2}\|_p & \cdots & \|A_{mq}\|_p \end{bmatrix} \right\|_p. \quad (2.2)$$

Further, the inequality becomes an equality if the F -norm is used.

Proof. It is obvious that if the F -norm is used, then the right hand side of inequality (2.2) equals the left hand side. Hence only the induced p -norm cases, $1 \leq p \leq \infty$, will be shown. Let a vector x be partitioned consistently with A as

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_q \end{bmatrix};$$

and note that

$$\|x\|_p = \left\| \begin{bmatrix} \|x_1\|_p \\ \|x_2\|_p \\ \vdots \\ \|x_q\|_p \end{bmatrix} \right\|_p.$$

Then

$$\begin{aligned} \| [A_{ij}] \|_p &:= \sup_{\|x\|_p=1} \| [A_{ij}] x \|_p = \sup_{\|x\|_p=1} \left\| \begin{bmatrix} \sum_{j=1}^q A_{1j} x_j \\ \sum_{j=1}^q A_{2j} x_j \\ \vdots \\ \sum_{j=1}^q A_{mj} x_j \end{bmatrix} \right\|_p \\ &= \sup_{\|x\|_p=1} \left\| \begin{bmatrix} \left\| \sum_{j=1}^q A_{1j} x_j \right\|_p \\ \left\| \sum_{j=1}^q A_{2j} x_j \right\|_p \\ \vdots \\ \left\| \sum_{j=1}^q A_{mj} x_j \right\|_p \end{bmatrix} \right\|_p \leq \sup_{\|x\|_p=1} \left\| \begin{bmatrix} \sum_{j=1}^q \|A_{1j}\|_p \|x_j\|_p \\ \sum_{j=1}^q \|A_{2j}\|_p \|x_j\|_p \\ \vdots \\ \sum_{j=1}^q \|A_{mj}\|_p \|x_j\|_p \end{bmatrix} \right\|_p \\ &= \sup_{\|x\|_p=1} \left\| \begin{bmatrix} \|A_{11}\|_p & \|A_{12}\|_p & \cdots & \|A_{1q}\|_p \\ \|A_{21}\|_p & \|A_{22}\|_p & \cdots & \|A_{2q}\|_p \\ \vdots & \vdots & & \vdots \\ \|A_{m1}\|_p & \|A_{m2}\|_p & \cdots & \|A_{mq}\|_p \end{bmatrix} \begin{bmatrix} \|x_1\|_p \\ \|x_2\|_p \\ \vdots \\ \|x_q\|_p \end{bmatrix} \right\|_p \\ &\leq \sup_{\|x\|_p=1} \left\| \left[\|A_{ij}\|_p \right] \|x\|_p \right\|_p \\ &= \left\| \left[\|A_{ij}\|_p \right] \right\|_p. \end{aligned}$$

□

2.8 Singular Value Decomposition

A very useful tool in matrix analysis is *Singular Value Decomposition (SVD)*. It will be seen that singular values of a matrix are good measures of the “size” of the matrix and that the corresponding singular vectors are good indications of strong/weak input or output directions.

Theorem 2.11 *Let $A \in \mathbb{F}^{m \times n}$. There exist unitary matrices*

$$\begin{aligned} U &= [u_1, u_2, \dots, u_m] \in \mathbb{F}^{m \times m} \\ V &= [v_1, v_2, \dots, v_n] \in \mathbb{F}^{n \times n} \end{aligned}$$

such that

$$A = U \Sigma V^*, \quad \Sigma = \begin{bmatrix} \Sigma_1 & 0 \\ 0 & 0 \end{bmatrix}$$

where

$$\Sigma_1 = \begin{bmatrix} \sigma_1 & 0 & \cdots & 0 \\ 0 & \sigma_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_p \end{bmatrix}$$

and

$$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_p \geq 0, \quad p = \min\{m, n\}.$$

Proof. Let $\sigma = \|A\|$ and without loss of generality assume $m \geq n$. Then from the definition of $\|A\|$, there exists a $z \in \mathbb{F}^n$ such that

$$\|Az\| = \sigma \|z\|.$$

By Lemma 2.8, there is a matrix $\tilde{U} \in \mathbb{F}^{m \times n}$ such that $\tilde{U}^* \tilde{U} = I$ and

$$Az = \sigma \tilde{U}z.$$

Now let

$$x = \frac{z}{\|z\|} \in \mathbb{F}^n, \quad y = \frac{\tilde{U}z}{\|\tilde{U}z\|} \in \mathbb{F}^m.$$

We have $Ax = \sigma y$. Let

$$V = [x \quad V_1] \in \mathbb{F}^{n \times n}$$

and

$$U = [y \quad U_1] \in \mathbb{F}^{m \times m}$$

be unitary.⁴ Consequently, U^*AV has the following structure:

$$A_1 := U^*AV = \begin{bmatrix} \sigma & w^* \\ 0 & B \end{bmatrix}$$

⁴Recall that it is always possible to extend an orthonormal set of vectors to an orthonormal basis for the whole space.

where $w \in \mathbb{F}^{n-1}$ and $B \in \mathbb{F}^{(m-1) \times (n-1)}$.

Since

$$\left\| A_1^* \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \right\|_2^2 = (\sigma^2 + w^*w),$$

it follows that $\|A_1\|^2 \geq \sigma^2 + w^*w$. But since $\sigma = \|A\| = \|A_1\|$, we must have $w = 0$. An obvious induction argument gives

$$U^*AV = \Sigma.$$

This completes the proof. $\square$

The σ_i is the i -th *singular value* of A , and the vectors u_i and v_i are, respectively, the i -th *left singular vector* and the i -th *right singular vector*. It is easy to verify that

$$\begin{aligned} Av_i &= \sigma_i u_i \\ A^*u_i &= \sigma_i v_i. \end{aligned}$$

The above equations can also be written as

$$\begin{aligned} A^*Av_i &= \sigma_i^2 v_i \\ AA^*u_i &= \sigma_i^2 u_i. \end{aligned}$$

Hence σ_i^2 is an eigenvalue of AA^* or A^*A , u_i is an eigenvector of AA^* , and v_i is an eigenvector of A^*A .

The following notations for singular values are often adopted:

$$\bar{\sigma}(A) = \sigma_{\max}(A) = \sigma_1 = \text{the largest singular value of } A;$$

and

$$\underline{\sigma}(A) = \sigma_{\min}(A) = \sigma_p = \text{the smallest singular value of } A.$$

Geometrically, the singular values of a matrix A are precisely the lengths of the semi-axes of the hyperellipsoid E defined by

$$E = \{y : y = Ax, x \in \mathbb{C}^n, \|x\| = 1\}.$$

Thus v_1 is the direction in which $\|y\|$ is largest for all $\|x\| = 1$; while v_n is the direction in which $\|y\|$ is smallest for all $\|x\| = 1$. From the input/output point of view, v_1 (v_n) is the *highest (lowest) gain input direction*, while u_1 (u_m) is the *highest (lowest) gain observing direction*. This can be illustrated by the following 2×2 matrix:

$$A = \begin{bmatrix} \cos \theta_1 & -\sin \theta_1 \\ \sin \theta_1 & \cos \theta_1 \end{bmatrix} \begin{bmatrix} \sigma_1 & \\ & \sigma_2 \end{bmatrix} \begin{bmatrix} \cos \theta_2 & -\sin \theta_2 \\ \sin \theta_2 & \cos \theta_2 \end{bmatrix}.$$

It is easy to see that A maps a unit disk to an ellipsoid with semi-axes of σ_1 and σ_2 .

Hence it is often convenient to introduce the following alternative definitions for the largest singular value $\bar{\sigma}$:

$$\bar{\sigma}(A) := \max_{\|x\|=1} \|Ax\|$$

and for the smallest singular value $\underline{\sigma}$ of a *tall matrix*:

$$\underline{\sigma}(A) := \min_{\|x\|=1} \|Ax\|.$$

Lemma 2.12 *Suppose A and Δ are square matrices. Then*

- (i) $|\underline{\sigma}(A + \Delta) - \underline{\sigma}(A)| \leq \bar{\sigma}(\Delta)$;
- (ii) $\underline{\sigma}(A\Delta) \geq \underline{\sigma}(A)\underline{\sigma}(\Delta)$;
- (iii) $\bar{\sigma}(A^{-1}) = \frac{1}{\underline{\sigma}(A)}$ if A is invertible.

Proof.

(i) By definition

$$\begin{aligned} \underline{\sigma}(A + \Delta) &:= \min_{\|x\|=1} \|(A + \Delta)x\| \\ &\geq \min_{\|x\|=1} \{\|Ax\| - \|\Delta x\|\} \\ &\geq \min_{\|x\|=1} \|Ax\| - \max_{\|x\|=1} \|\Delta x\| \\ &= \underline{\sigma}(A) - \bar{\sigma}(\Delta). \end{aligned}$$

Hence $-\bar{\sigma}(\Delta) \leq \underline{\sigma}(A + \Delta) - \underline{\sigma}(A)$. The other inequality $\underline{\sigma}(A + \Delta) - \underline{\sigma}(A) \leq \bar{\sigma}(\Delta)$ follows by replacing A by $A + \Delta$ and Δ by $-\Delta$ in the above proof.

(ii) This follows by noting that

$$\begin{aligned} \underline{\sigma}(A\Delta) &:= \min_{\|x\|=1} \|A\Delta x\| \\ &= \sqrt{\min_{\|x\|=1} x^* \Delta^* A^* A \Delta x} \\ &\geq \underline{\sigma}(A) \min_{\|x\|=1} \|\Delta x\| \\ &= \underline{\sigma}(A)\underline{\sigma}(\Delta). \end{aligned}$$

- (iii) Let the singular value decomposition of A be $A = U\Sigma V^*$, then $A^{-1} = V\Sigma^{-1}U^*$. Hence $\bar{\sigma}(A^{-1}) = \bar{\sigma}(\Sigma^{-1}) = 1/\underline{\sigma}(\Sigma) = 1/\underline{\sigma}(A)$.

□

Some useful properties of SVD are collected in the following lemma.

Lemma 2.13 *Let $A \in \mathbb{F}^{m \times n}$ and*

$$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > \sigma_{r+1} = \cdots = 0, \quad r \leq \min\{m, n\}.$$

Then

1. $\text{rank}(A) = r$;
2. $\text{Ker}A = \text{span}\{v_{r+1}, \dots, v_n\}$ and $(\text{Ker}A)^\perp = \text{span}\{v_1, \dots, v_r\}$;
3. $\text{Im}A = \text{span}\{u_1, \dots, u_r\}$ and $(\text{Im}A)^\perp = \text{span}\{u_{r+1}, \dots, u_m\}$;
4. $A \in \mathbb{F}^{m \times n}$ has a dyadic expansion:
$$A = \sum_{i=1}^r \sigma_i u_i v_i^* = U_r \Sigma_r V_r^*$$

where $U_r = [u_1, \dots, u_r]$, $V_r = [v_1, \dots, v_r]$, and $\Sigma_r = \text{diag}(\sigma_1, \dots, \sigma_r)$;
5. $\|A\|_F^2 = \sigma_1^2 + \sigma_2^2 + \cdots + \sigma_r^2$;
6. $\|A\| = \sigma_1$;
7. $\sigma_i(U_0 A V_0) = \sigma_i(A)$, $i = 1, \dots, p$ for any appropriately dimensioned unitary matrices U_0 and V_0 ;
8. Let $k < r = \text{rank}(A)$ and $A_k := \sum_{i=1}^k \sigma_i u_i v_i^*$, then

$$\min_{\text{rank}(B) \leq k} \|A - B\| = \|A - A_k\| = \sigma_{k+1}.$$

Proof. We shall only give a proof for part 8. It is easy to see that $\text{rank}(A_k) \leq k$ and $\|A - A_k\| = \sigma_{k+1}$. Hence, we only need to show that $\min_{\text{rank}(B) \leq k} \|A - B\| \geq \sigma_{k+1}$. Let

B be any matrix such that $\text{rank}(B) \leq k$. Then

$$\begin{aligned} \|A - B\| &= \|U \Sigma V^* - B\| = \|\Sigma - U^* B V\| \\ &\geq \left\| \begin{bmatrix} I_{k+1} & 0 \end{bmatrix} (\Sigma - U^* B V) \begin{bmatrix} I_{k+1} \\ 0 \end{bmatrix} \right\| = \|\Sigma_{k+1} - \hat{B}\| \end{aligned}$$

where $\hat{B} = \begin{bmatrix} I_{k+1} & 0 \end{bmatrix} U^* B V \begin{bmatrix} I_{k+1} \\ 0 \end{bmatrix} \in \mathbb{F}^{(k+1) \times (k+1)}$ and $\text{rank}(\hat{B}) \leq k$. Let $x \in \mathbb{F}^{k+1}$ be such that $\hat{B}x = 0$ and $\|x\| = 1$. Then

$$\|A - B\| \geq \|\Sigma_{k+1} - \hat{B}\| \geq \|(\Sigma_{k+1} - \hat{B})x\| = \|\Sigma_{k+1}x\| \geq \sigma_{k+1}.$$

Since B is arbitrary, the conclusion follows. □

2.9 Generalized Inverses

Let $A \in \mathbb{C}^{m \times n}$. A matrix $X \in \mathbb{C}^{n \times m}$ is said to be a *right inverse* of A if $AX = I$. Obviously, A has a right inverse iff A has full row rank, and, in that case, one of the right inverses is given by $X = A^*(AA^*)^{-1}$. Similarly, if $YA = I$ then Y is called a *left inverse* of A . By duality, A has a left inverse iff A has full column rank, and, furthermore, one of the left inverses is $Y = (A^*A)^{-1}A^*$. Note that right (or left) inverses are not necessarily unique. For example, any matrix in the form $\begin{bmatrix} I \\ \star \end{bmatrix}$ is a right inverse of $\begin{bmatrix} I & 0 \end{bmatrix}$.

More generally, if a matrix A has neither full row rank nor full column rank, then all the ordinary matrix inverses do not exist; however, the so called *pseudo-inverse*, known also as the *Moore-Penrose inverse*, is useful. This pseudo-inverse is denoted by A^+ , which satisfies the following conditions:

- (i) $AA^+A = A$;
- (ii) $A^+AA^+ = A^+$;
- (iii) $(AA^+)^* = AA^+$;
- (iv) $(A^+A)^* = A^+A$.

It can be shown that pseudo-inverse is unique. One way of computing A^+ is by writing

$$A = BC$$

so that B has full column rank and C has full row rank. Then

$$A^+ = C^*(CC^*)^{-1}(B^*B)^{-1}B^*.$$

Another way to compute A^+ is by using SVD. Suppose A has a singular value decomposition

$$A = U\Sigma V^*$$

with

$$\Sigma = \begin{bmatrix} \Sigma_r & 0 \\ 0 & 0 \end{bmatrix}, \quad \Sigma_r > 0.$$

Then $A^+ = V\Sigma^+U^*$ with

$$\Sigma^+ = \begin{bmatrix} \Sigma_r^{-1} & 0 \\ 0 & 0 \end{bmatrix}.$$

2.10 Semidefinite Matrices

A square hermitian matrix $A = A^*$ is said to be *positive definite (semi-definite)*, denoted by $A > 0$ (≥ 0), if $x^*Ax > 0$ (≥ 0) for all $x \neq 0$. Suppose $A \in \mathbb{F}^{n \times n}$ and $A = A^* \geq 0$, then there exists a $B \in \mathbb{F}^{n \times r}$ with $r \geq \text{rank}(A)$ such that $A = BB^*$.

Lemma 2.14 *Let $B \in \mathbb{F}^{m \times n}$ and $C \in \mathbb{F}^{k \times n}$. Suppose $m \geq k$ and $B^*B = C^*C$. Then there exists a matrix $U \in \mathbb{F}^{m \times k}$ such that $U^*U = I$ and $B = UC$.*

Proof. Let V_1 and V_2 be unitary matrices such that

$$B_1 = V_1 \begin{bmatrix} B_1 \\ 0 \end{bmatrix}, \quad C_1 = V_2 \begin{bmatrix} C_1 \\ 0 \end{bmatrix}$$

where B_1 and C_1 are full row rank. Then B_1 and C_1 have the same number of rows and $V_3 := B_1 C_1^* (C_1 C_1^*)^{-1}$ satisfies $V_3^* V_3 = I$ since $B^* B = C^* C$. Hence V_3 is a unitary matrix and $V_3^* B_1 = C_1$. Finally let

$$U = V_1 \begin{bmatrix} V_3 & 0 \\ 0 & V_4 \end{bmatrix} V_2^*$$

for any suitably dimensioned V_4 such that $V_4^* V_4 = I$. $\square$

We can define square root for a positive semi-definite matrix A , $A^{1/2} = (A^{1/2})^* \geq 0$, by

$$A = A^{1/2} A^{1/2}.$$

Clearly, $A^{1/2}$ can be computed by using spectral decomposition or SVD: let $A = U \Lambda U^*$, then

$$A^{1/2} = U \Lambda^{1/2} U^*$$

where

$$\Lambda = \text{diag}\{\lambda_1, \dots, \lambda_n\}, \quad \Lambda^{1/2} = \text{diag}\{\sqrt{\lambda_1}, \dots, \sqrt{\lambda_n}\}.$$

Lemma 2.15 Suppose $A = A^* > 0$ and $B = B^* \geq 0$. Then $A > B$ iff $\rho(BA^{-1}) < 1$.

Proof. Since $A > 0$, we have $A > B$ iff

$$0 < I - A^{-1/2} B A^{-1/2} = I - A^{-1/2} (B A^{-1}) A^{1/2}.$$

However, $A^{-1/2} B A^{-1/2}$ and $B A^{-1}$ are similar, and hence $\lambda_i(B A^{-1}) = \lambda_i(A^{-1/2} B A^{-1/2})$. Therefore, the conclusion follows by the fact that

$$0 < I - A^{-1/2} B A^{-1/2}$$

iff $\rho(A^{-1/2} B A^{-1/2}) < 1$ iff $\rho(B A^{-1}) < 1$. $\square$

Lemma 2.16 Let $X = X^* \geq 0$ be partitioned as

$$X = \begin{bmatrix} X_{11} & X_{12} \\ X_{12}^* & X_{22} \end{bmatrix}.$$

Then $\text{Ker} X_{22} \subset \text{Ker} X_{12}$. Consequently, if X_{22}^+ is the pseudo-inverse of X_{22} , then $Y = X_{12} X_{22}^+$ solves

$$Y X_{22} = X_{12}$$

and

$$\begin{bmatrix} X_{11} & X_{12} \\ X_{12}^* & X_{22} \end{bmatrix} = \begin{bmatrix} I & X_{12} X_{22}^+ \\ 0 & I \end{bmatrix} \begin{bmatrix} X_{11} - X_{12} X_{22}^+ X_{12}^* & 0 \\ 0 & X_{22} \end{bmatrix} \begin{bmatrix} I & 0 \\ X_{22}^+ X_{12}^* & I \end{bmatrix}.$$

Proof. Without loss of generality, assume

$$X_{22} = \begin{bmatrix} U_1 & U_2 \end{bmatrix} \begin{bmatrix} \Sigma_1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} U_1^* \\ U_2^* \end{bmatrix}$$

with $\Sigma_1 = \Sigma_1^* > 0$ and $U = \begin{bmatrix} U_1 & U_2 \end{bmatrix}$ unitary. Then

$$\text{Ker } X_{22} = \text{span}\{\text{columns of } U_2\}$$

and

$$X_{22}^+ = \begin{bmatrix} U_1 & U_2 \end{bmatrix} \begin{bmatrix} \Sigma_1^{-1} & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} U_1^* \\ U_2^* \end{bmatrix}.$$

Moreover

$$\begin{bmatrix} I & 0 \\ 0 & U^* \end{bmatrix} \begin{bmatrix} X_{11} & X_{12} \\ X_{12}^* & X_{22} \end{bmatrix} \begin{bmatrix} I & 0 \\ 0 & U \end{bmatrix} \geq 0$$

gives $X_{12}U_2 = 0$. Hence, $\text{Ker } X_{22} \subset \text{Ker } X_{12}$ and now

$$X_{12}X_{22}^+X_{22} = X_{12}U_1U_1^* = X_{12}U_1U_1^* + X_{12}U_2U_2^* = X_{12}.$$

The factorization follows easily. $\square$

2.11 Matrix Dilation Problems*

In this section, we consider the following induced 2-norm optimization problem:

$$\min_X \left\| \begin{bmatrix} X & B \\ C & A \end{bmatrix} \right\| \quad (2.3)$$

where X , B , C , and A are constant matrices of compatible dimensions.

The matrix $\begin{bmatrix} X & B \\ C & A \end{bmatrix}$ is a *dilation* of its sub-matrices as indicated in the following diagram:

$$\begin{array}{ccc} \begin{bmatrix} X & B \\ C & A \end{bmatrix} & \begin{array}{c} \xleftarrow{d} \\ \xrightarrow{c} \end{array} & \begin{bmatrix} B \\ A \end{bmatrix} \\ \begin{array}{c} \uparrow d \\ \downarrow c \end{array} & & \begin{array}{c} \uparrow d \\ \downarrow c \end{array} \\ \begin{bmatrix} C & A \end{bmatrix} & \begin{array}{c} \xleftarrow{d} \\ \xrightarrow{c} \end{array} & \begin{bmatrix} A \end{bmatrix} \end{array}$$

In this diagram, “*c*” stands for the operation of *compression* and “*d*” stands for *dilation*. Compression is always norm non-increasing and dilation is always norm non-decreasing. Sometimes dilation can be made to be norm preserving. Norm preserving dilations are the focus of this section.

The simplest matrix dilation problem occurs when solving

$$\min_X \left\| \begin{bmatrix} X \\ A \end{bmatrix} \right\|. \quad (2.4)$$

Although (2.4) is a much simplified version of (2.3), we will see that it contains all the essential features of the general problem. Letting γ_0 denote the minimum norm in (2.4), it is immediate that

$$\gamma_0 = \|A\|.$$

The following theorem characterizes all solutions to (2.4).

Theorem 2.17 $\forall \gamma \geq \gamma_0$,

$$\left\| \begin{bmatrix} X \\ A \end{bmatrix} \right\| \leq \gamma$$

iff there is a Y with $\|Y\| \leq 1$ such that

$$X = Y(\gamma^2 I - A^* A)^{1/2}.$$

Proof.

$$\left\| \begin{bmatrix} X \\ A \end{bmatrix} \right\| \leq \gamma$$

iff

$$X^* X + A^* A \leq \gamma^2 I$$

iff

$$X^* X \leq (\gamma^2 I - A^* A).$$

Now suppose $X^* X \leq (\gamma^2 I - A^* A)$ and let

$$Y := X \left[(\gamma^2 I - A^* A)^{1/2} \right]^+$$

then $X = Y(\gamma^2 I - A^* A)^{1/2}$ and $Y^* Y \leq I$. Similarly if $X = Y(\gamma^2 I - A^* A)^{1/2}$ and $Y^* Y \leq I$ then $X^* X \leq (\gamma^2 I - A^* A)$. $\square$

This theorem implies that, in general, (2.4) has more than one solution, which is in contrast to the minimization in the Frobenius norm in which $X = 0$ is the unique solution. The solution $X = 0$ is the central solution but others are possible unless $A^* A = \gamma_0^2 I$.

Remark 2.1 The theorem still holds if $(\gamma^2 I - A^* A)^{1/2}$ is replaced by any matrix R such that $\gamma^2 I - A^* A = R^* R$. $\heartsuit$

A more restricted version of the above theorem is shown in the following corollary.

Corollary 2.18 $\forall \gamma > \gamma_0$,

$$\left\| \begin{bmatrix} X \\ A \end{bmatrix} \right\| \leq \gamma (< \gamma)$$

iff

$$\left\| X(\gamma^2 I - A^* A)^{-1/2} \right\| \leq 1 (< 1).$$

The corresponding dual results are

Theorem 2.19 $\forall \gamma \geq \gamma_0$

$$\left\| \begin{bmatrix} X & A \end{bmatrix} \right\| \leq \gamma$$

iff there is a Y , $\|Y\| \leq 1$, such that

$$X = (\gamma^2 I - AA^*)^{1/2} Y.$$

Corollary 2.20 $\forall \gamma > \gamma_0$

$$\left\| \begin{bmatrix} X & A \end{bmatrix} \right\| \leq \gamma (< \gamma)$$

iff

$$\left\| (\gamma^2 I - AA^*)^{-1/2} X \right\| \leq 1 (< 1).$$

Now, returning to the problem in (2.3), let

$$\gamma_0 := \min_X \left\| \begin{bmatrix} X & B \\ C & A \end{bmatrix} \right\|. \quad (2.5)$$

The following so called Parrott's theorem will play an important role in many control related optimization problems. The proof is the straightforward application of Theorem 2.17 and its dual, Theorem 2.19.

Theorem 2.21 (Parrott's Theorem) *The minimum in (2.5) is given by*

$$\gamma_0 = \max \left\{ \left\| \begin{bmatrix} C & A \end{bmatrix} \right\|, \left\| \begin{bmatrix} B \\ A \end{bmatrix} \right\| \right\}. \quad (2.6)$$

Proof. Denote by $\hat{\gamma}$ the right hand side of the equation (2.6). Clearly, $\gamma_0 \geq \hat{\gamma}$ since compressions are norm non-increasing, and that $\gamma_0 \leq \hat{\gamma}$ will be shown by using Theorem 2.17 and Theorem 2.19.

Suppose $A \in \mathbb{C}^{n \times m}$ and $n \geq m$ (the case for $m > n$ can be shown in the same fashion). Then A has the following singular value decomposition:

$$A = U \begin{bmatrix} \Sigma_m \\ 0_{n-m, m} \end{bmatrix} V^*, \quad U \in \mathbb{C}^{n \times n}, V \in \mathbb{C}^{m \times m}.$$

Hence

$$\hat{\gamma}^2 I - A^* A = V(\hat{\gamma}^2 I - \Sigma_m^2) V^*$$

and

$$\hat{\gamma}^2 I - AA^* = U \begin{bmatrix} \hat{\gamma}^2 I - \Sigma_m^2 & 0 \\ 0 & \hat{\gamma}^2 I_{n-m} \end{bmatrix} U^*.$$

Now let

$$(\hat{\gamma}^2 I - A^* A)^{1/2} := V(\hat{\gamma}^2 I - \Sigma_m^2)^{1/2} V^*$$

and

$$(\hat{\gamma}^2 I - AA^*)^{1/2} := U \begin{bmatrix} (\hat{\gamma}^2 I - \Sigma_m^2)^{1/2} & 0 \\ 0 & \hat{\gamma} I_{n-m} \end{bmatrix} U^*.$$

Then it is easy to verify that

$$(\hat{\gamma}^2 I - A^* A)^{1/2} A^* = A^* (\hat{\gamma}^2 I - AA^*)^{1/2}.$$

Using this equality, we can show that

$$\begin{aligned} & \begin{bmatrix} -A^* & (\hat{\gamma}^2 I - A^* A)^{1/2} \\ (\hat{\gamma}^2 I - AA^*)^{1/2} & A \end{bmatrix} \begin{bmatrix} -A^* & (\hat{\gamma}^2 I - A^* A)^{1/2} \\ (\hat{\gamma}^2 I - AA^*)^{1/2} & A \end{bmatrix}^* \\ &= \begin{bmatrix} \hat{\gamma}^2 I & 0 \\ 0 & \hat{\gamma}^2 I \end{bmatrix}. \end{aligned}$$

Now we are ready to show that $\gamma_0 \leq \hat{\gamma}$.

From Theorem 2.17 we have that $B = Y(\hat{\gamma}^2 I - A^* A)^{1/2}$ for some Y such that $\|Y\| \leq 1$. Similarly, Theorem 2.19 yields $C = (\hat{\gamma}^2 I - AA^*)^{1/2} Z$ for some Z with $\|Z\| \leq 1$. Now let $\hat{X} = -YA^*Z$. Then

$$\begin{aligned} \left\| \begin{bmatrix} \hat{X} & B \\ C & A \end{bmatrix} \right\| &= \left\| \begin{bmatrix} -YA^*Z & Y(\hat{\gamma}^2 I - A^* A)^{1/2} \\ (\hat{\gamma}^2 I - AA^*)^{1/2} Z & A \end{bmatrix} \right\| \\ &= \left\| \begin{bmatrix} Y & \\ & I \end{bmatrix} \begin{bmatrix} -A^* & (\hat{\gamma}^2 I - A^* A)^{1/2} \\ (\hat{\gamma}^2 I - AA^*)^{1/2} & A \end{bmatrix} \begin{bmatrix} Z & \\ & I \end{bmatrix} \right\| \\ &\leq \left\| \begin{bmatrix} -A^* & (\hat{\gamma}^2 I - A^* A)^{1/2} \\ (\hat{\gamma}^2 I - AA^*)^{1/2} & A \end{bmatrix} \right\| \\ &= \hat{\gamma}. \end{aligned}$$

Thus $\hat{\gamma} \geq \gamma_0$, so $\hat{\gamma} = \gamma_0$. $\square$

This theorem gives one solution to (2.3) and an expression for γ_0 . As in (2.4), there may be more than one solution to (2.3), although the proof of theorem 2.21 exhibits only one. Theorem 2.22 considers the problem of parameterizing all solutions. The solution $\hat{X} = -YA^*Z$ is the “central” solution analogous to $X = 0$ in (2.4).

Theorem 2.22 *Suppose $\gamma \geq \gamma_0$. The solutions X such that*

$$\left\| \begin{bmatrix} X & B \\ C & A \end{bmatrix} \right\| \leq \gamma \quad (2.7)$$

are exactly those of the form

$$X = -YA^*Z + \gamma(I - YY^*)^{1/2}W(I - Z^*Z)^{1/2} \quad (2.8)$$

where W is an arbitrary contraction ($\|W\| \leq 1$), and Y with $\|Y\| \leq 1$ and Z with $\|Z\| \leq 1$ solve the linear equations

$$B = Y(\gamma^2 I - A^*A)^{1/2}, \quad (2.9)$$

$$C = (\gamma^2 I - AA^*)^{1/2}Z. \quad (2.10)$$

Proof. Since $\gamma \geq \gamma_0$, again from Theorem 2.19 there exists a Z with $\|Z\| \leq 1$ such that

$$C = (\gamma^2 I - AA^*)^{1/2}Z.$$

Note that using the above expression for C we have

$$\begin{aligned} & \gamma^2 I - \begin{bmatrix} C & A \end{bmatrix}^* \begin{bmatrix} C & A \end{bmatrix} \\ &= \begin{bmatrix} \gamma(I - Z^*Z)^{1/2} & 0 \\ -A^*Z & (\gamma^2 I - A^*A)^{1/2} \end{bmatrix}^* \begin{bmatrix} \gamma(I - Z^*Z)^{1/2} & 0 \\ -A^*Z & (\gamma^2 I - A^*A)^{1/2} \end{bmatrix}. \end{aligned}$$

Now apply Theorem 2.17 (Remark 2.1) to inequality (2.7) with respect to the partitioned matrix $\begin{bmatrix} X & B \\ C & A \end{bmatrix}$ to get

$$\begin{bmatrix} X & B \end{bmatrix} = \hat{W} \begin{bmatrix} \gamma(I - Z^*Z)^{1/2} & 0 \\ -A^*Z & (\gamma^2 I - A^*A)^{1/2} \end{bmatrix}$$

for some contraction $\hat{W}$, $\|\hat{W}\| \leq 1$. Partition $\hat{W}$ as $\hat{W} = \begin{bmatrix} W_1 & Y \end{bmatrix}$ to obtain the expression for X and B :

$$\begin{aligned} X &= -YA^*Z + \gamma W_1(I - Z^*Z)^{1/2}, \\ B &= Y(\gamma^2 I - A^*A)^{1/2}. \end{aligned}$$

Then $\|Y\| \leq 1$ and the theorem follows by noting that $\|\begin{bmatrix} W_1 & Y \end{bmatrix}\| \leq 1$ iff there is a W , $\|W\| \leq 1$, such that $W_1 = (I - YY^*)^{1/2}W$. $\square$

The following corollary gives an alternative version of Theorem 2.22 when $\gamma > \gamma_0$.

Corollary 2.23 For $\gamma > \gamma_0$.

$$\left\| \begin{bmatrix} X & B \\ C & A \end{bmatrix} \right\| \leq \gamma \quad (< \gamma) \quad (2.11)$$

iff

$$\left\| (I - YY^*)^{-1/2} (X + YA^*Z) (I - Z^*Z)^{-1/2} \right\| \leq \gamma \quad (< \gamma) \quad (2.12)$$

where

$$Y = B(\gamma^2 I - A^*A)^{-1/2}, \quad (2.13)$$

$$Z = (\gamma^2 I - AA^*)^{-1/2}C. \quad (2.14)$$

Note that in the case of $\gamma > \gamma_0$, $I - YY^*$ and $I - Z^*Z$ are invertible since Corollary 2.18 and 2.20 clearly show that $\|Y\| < 1$ and $\|Z\| < 1$. There are many alternative characterizations of solutions to (2.11), although the formulas given above seem to be the simplest.

As a straightforward application of the dilation results obtained above, consider the following matrix approximation problem:

$$\gamma_0 = \min_Q \|R + UQV\| \quad (2.15)$$

where R , U , and V are constant matrices such that $U^*U = I$ and $VV^* = I$.

Corollary 2.24 The minimum achievable norm is given by

$$\gamma_0 = \max \{ \|U_\perp^* R\|, \|RV_\perp^*\| \},$$

and the parameterization for all optimal solutions Q can be obtained from Theorem 2.22 with $X = Q + U^*RV^*$, $A = U_\perp^*RV_\perp^*$, $B = U^*RV_\perp^*$, and $C = U_\perp^*RV^*$.

Proof. Let $U_\perp$ and $V_\perp$ be such that $\begin{bmatrix} U & U_\perp \end{bmatrix}$ and $\begin{bmatrix} V \\ V_\perp \end{bmatrix}$ are unitary matrices. Then

$$\begin{aligned} \gamma_0 &= \min_Q \|R + UQV\| \\ &= \min_Q \left\| \begin{bmatrix} U & U_\perp \end{bmatrix}^* (R + UQV) \begin{bmatrix} V \\ V_\perp \end{bmatrix}^* \right\| \\ &= \min_Q \left\| \begin{bmatrix} U^*RV^* + Q & U^*RV_\perp^* \\ U_\perp^*RV^* & U_\perp^*RV_\perp^* \end{bmatrix} \right\|. \end{aligned}$$

The result follows after applying Theorem 2.21 and 2.22. $\square$

A similar problem arises in $\mathcal{H}_\infty$ control theory.

2.12 Notes and References

A very extensive treatment of most topics in this chapter can be found in Brogan [1991], Horn and Johnson [1990,1991] and Lancaster and Tismenetsky [1985]. Golub and Van Loan's book [1983] contains many numerical algorithms for solving most of the problems in this chapter. The matrix dilation theory can be found in Davis, Kahan, and Weinberger [1982].